Consulting Project: NurseNext Scheduling Optimization

Management Science - Healthcare Workforce Management



Client Briefing: NurseNext Hospital

Meet Your Client

- **Industry:** Healthcare
- **Client Contact:** Dr. Robin Torres, Chief Operating Officer
- **Facility:** Medium-sized hospital with 3 departments
- **Staff:** 20 nurses, 50 beds, 24/7 operations

The COO's Challenge

Dr. Robin Torres, COO of NurseNext Hospital

"I became a doctor to help people, but now I spend 8 hours every week manually creating nurse schedules and they're still terrible!

The situation is unsustainable:

- *Massive overtime costs*
- *25% sick leave rate because nurses are exhausted*
- *High turnover, we lose 3-4 experienced nurses annually*
- *Constant complaints about unfair weekend distribution*
- *Last month, 2 nurses quit citing 'burnout and unfair scheduling'*

The manual process is killing us. I try to be fair, but with 20 nurses, 3 departments, different skill levels, shift preferences, and labor laws... I just can't optimize it all in my head.

We need an automated scheduling system that:

1. *Cuts overtime massively, if possible*
2. *Distributes weekends fairly (the biggest complaint)*
3. *Respects labor laws (consecutive shifts, rest periods)*
4. *Maintains quality of care (right skills at right times)*
5. *Is robust to sick calls (we always have 1-2 nurses out)*

Can you build this for us? I have data from last month's schedules and our staffing requirements."

The Business Context

Current Situation

Departments:

- **Emergency Department (ED):** High-intensity, requires experienced nurses

- **Medical-Surgical (Med-Surg):** General care, moderate intensity
- **Intensive Care Unit (ICU):** Critical patients, highest skill requirements

Staffing Model:

NurseNext uses a modern **hybrid staffing model**:

- **Department Specialists (13 nurses):** Core staff assigned to specific departments
  - 4 ED specialists, 5 Med-Surg specialists, 4 ICU specialists
  - Deep expertise in their department's specialized care
  - Only work in their assigned department
- **Flexible Float Pool (7 nurses):** Cross-trained resource nurses
  - Can work in **any** department (ED, Med-Surg, or ICU)
  - Provide flexibility for workload balancing across departments
  - Essential for covering peak demand and sick calls

This creates an interesting **resource allocation challenge**: How should flexible nurses be distributed across departments to minimize cost and maximize fairness?

Shifts (Each 8 hours):

- **Morning (7:00-15:00):** Highest patient activity
- **Evening (15:00-23:00):** Moderate activity
- **Night (23:00-7:00):** Lower activity but critical monitoring

Scheduling Horizon: 1 week (21 shifts total = 7 days × 3 shifts/day)

Cost Structure

- **Regular wage:** €25/hour (€200 per 8-hour shift)
- **Overtime wage:** €37.50/hour (€300 per overtime shift, 1.5x multiplier)
- **Weekend premium:** +€50 per weekend shift
- **Night premium:** +€30 per night shift
- **Understaffing penalty:** €500 per understaffed shift (patient safety risk)

Labor Law Constraints

> ⚠ **Legal Requirements (Must Comply!)**
>
> 1. **Maximum consecutive shifts:** 5 days in a row
> 2. **Minimum rest period:** 11 hours between shifts
> 3. **Maximum weekly hours:** 48 hours per nurse (6 shifts max)
> 4. **Weekend work limit:** Maximum 2 weekend days per month

> ℹ **Note**
>
> **Important Note on Night Shifts:** Night shifts (23:00-07:00) end at 7:00 AM the *following calendar day*. Therefore, a nurse working Monday Night (ending Tuesday 7:00 AM) **cannot** work Tuesday Morning (0 hours rest) or Tuesday Evening (only 8 hours rest). Both violate the 11-hour minimum rest period.

Fairness Expectations

Nurses' union has negotiated fairness requirements:

- Weekend shifts distributed equally across nurses

- Night shifts distributed equally
- Workload (total hours) balanced within ±10%
- No nurse should work significantly more undesirable shifts than others

The Data

> **Scheduling Context**
> This schedule is for **Week 3 of the current month**. The `weekend_days_worked_this_month` field shows how many weekend days each nurse has already worked in Weeks 1-2. Since the monthly limit is 2 weekend days per month, nurses may have 0, 1, or 2 weekend days already worked, which constrains how many more weekend shifts they can take in this week's schedule.

Nurse Information

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from datetime import datetime, timedelta

# Set random seed for reproducibility
np.random.seed(2025)

# Generate nurse data
def generate_nurse_data():
    nurses = []

    # ED Specialists (4 nurses) - only work in ED
    for i in range(1, 5):
        nurses.append({
            'nurse_id': i,
            'name': f'Nurse ED-{i}',
            'department': 'ED',
            'departments_qualified': ['ED'],
            'nurse_type': 'Specialist',
            'skill_level': np.random.choice(['Junior', 'Senior', 'Senior',
'Expert'], p=[0.2, 0.5, 0.2, 0.1]),
            'experience_years': np.random.randint(1, 15),
            'can_work_night': np.random.choice([True, True, False], p=[0.6,
0.3, 0.1]),
            'weekend_days_worked_this_month': np.random.randint(0, 3)
        })

    # Med-Surg Specialists (5 nurses) - only work in Med-Surg
    for i in range(5, 10):
        nurses.append({
            'nurse_id': i,
            'name': f'Nurse MS-{i-4}',
```

```
            'department': 'Med-Surg',
            'departments_qualified': ['Med-Surg'],
            'nurse_type': 'Specialist',
            'skill_level': np.random.choice(['Junior', 'Senior', 'Senior'],
p=[0.3, 0.5, 0.2]),
            'experience_years': np.random.randint(1, 12),
            'can_work_night': np.random.choice([True, True, False], p=[0.7,
0.2, 0.1]),
            'weekend_days_worked_this_month': np.random.randint(0, 3)
        })

    # ICU Specialists (4 nurses) - only work in ICU
    for i in range(10, 14):
        nurses.append({
            'nurse_id': i,
            'name': f'Nurse ICU-{i-9}',
            'department': 'ICU',
            'departments_qualified': ['ICU'],
            'nurse_type': 'Specialist',
            'skill_level': np.random.choice(['Senior', 'Expert', 'Expert'],
p=[0.4, 0.4, 0.2]),
            'experience_years': np.random.randint(3, 20),
            'can_work_night': True,  # ICU nurses must be able to work nights
            'weekend_days_worked_this_month': np.random.randint(0, 3)
        })

    # Flexible Float Pool (7 nurses) - can work in ANY department
    for i in range(14, 21):
        nurses.append({
            'nurse_id': i,
            'name': f'Nurse FLOAT-{i-13}',
            'department': 'Float',
            'departments_qualified': ['ED', 'Med-Surg', 'ICU'],
            'nurse_type': 'Flexible',
            'skill_level': np.random.choice(['Junior', 'Senior', 'Senior',
'Expert'], p=[0.15, 0.5, 0.25, 0.1]),
            'experience_years': np.random.randint(2, 15),
            'can_work_night': np.random.choice([True, True, True, False],
p=[0.75, 0.15, 0.05, 0.05]),  # Most can work nights
            'weekend_days_worked_this_month': np.random.randint(0, 3)
        })

    return pd.DataFrame(nurses)

nurses_df = generate_nurse_data()

print("NURSENEXT HOSPITAL - NURSE ROSTER")
print("=" * 100)
```

```
print(nurses_df.to_string(index=False))
print("\n" + "=" * 100)
print(f"Total nurses: {len(nurses_df)}")
print(f"\nBy type:")
print(f"  Specialists: {len(nurses_df[nurses_df['nurse_type']=='Specialist'])} nurses")
print(f"    - ED: {len(nurses_df[nurses_df['department']=='ED'])} nurses")
print(f"    - Med-Surg: {len(nurses_df[nurses_df['department']=='Med-Surg'])} nurses")
print(f"    - ICU: {len(nurses_df[nurses_df['department']=='ICU'])} nurses")
print(f"  Flexible (Float Pool): {len(nurses_df[nurses_df['nurse_type']=='Flexible'])} nurses (can work any department)")
print(f"\nBy skill level:
Junior={len(nurses_df[nurses_df['skill_level']=='Junior'])}, "
      f"Senior={len(nurses_df[nurses_df['skill_level']=='Senior'])}, "
      f"Expert={len(nurses_df[nurses_df['skill_level']=='Expert'])}")
```

```
NURSENEXT HOSPITAL - NURSE ROSTER
====================================================================================================
 nurse_id          name department departments_qualified nurse_type skill_level experience_years  can_work_night  weekend_days_worked_this_month
        1    Nurse ED-1         ED                 [ED] Specialist      Junior                3            True                               0
        2    Nurse ED-2         ED                 [ED] Specialist      Senior               13           False                               0
        3    Nurse ED-3         ED                 [ED] Specialist      Senior                2           False                               1
        4    Nurse ED-4         ED                 [ED] Specialist      Senior                1            True                               0
        5    Nurse MS-1   Med-Surg           [Med-Surg] Specialist      Senior                1            True                               1
        6    Nurse MS-2   Med-Surg           [Med-Surg] Specialist      Senior                2            True                               1
        7    Nurse MS-3   Med-Surg           [Med-Surg] Specialist      Junior               10           False                               0
        8    Nurse MS-4   Med-Surg           [Med-Surg] Specialist      Senior                6           False                               0
        9    Nurse MS-5   Med-Surg           [Med-Surg] Specialist      Senior               10            True                               0
       10   Nurse ICU-1        ICU                [ICU] Specialist      Senior                8            True                               2
       11   Nurse ICU-2        ICU                [ICU] Specialist      Expert               10            True                               2
       12   Nurse ICU-3        ICU                [ICU] Specialist      Expert               18            True                               2
       13   Nurse ICU-4        ICU                [ICU] Specialist      Expert               18            True                               1
```

```
       14 Nurse FLOAT-1      Float   [ED, Med-Surg, ICU]   Flexible      Senior
4            True                                2
       15 Nurse FLOAT-2      Float   [ED, Med-Surg, ICU]   Flexible      Senior
2            True                                0
       16 Nurse FLOAT-3      Float   [ED, Med-Surg, ICU]   Flexible      Expert
8            True                                0
       17 Nurse FLOAT-4      Float   [ED, Med-Surg, ICU]   Flexible      Senior
5            True                                2
       18 Nurse FLOAT-5      Float   [ED, Med-Surg, ICU]   Flexible      Senior
2            True                                1
       19 Nurse FLOAT-6      Float   [ED, Med-Surg, ICU]   Flexible      Expert
11            True                               0
       20 Nurse FLOAT-7      Float   [ED, Med-Surg, ICU]   Flexible      Senior
8            True                                1

================================================================================================
Total nurses: 20

By type:
  Specialists: 13 nurses
    - ED: 4 nurses
    - Med-Surg: 5 nurses
    - ICU: 4 nurses
  Flexible (Float Pool): 7 nurses (can work any department)

By skill level: Junior=2, Senior=13, Expert=5
```

Shift Requirements

```
# Define shift requirements for each department and shift type
# Format: (min_nurses, min_senior_nurses)

shift_requirements = {
    'ED': {
        'Morning': {'min_nurses': 2, 'min_senior': 1},
        'Evening': {'min_nurses': 2, 'min_senior': 1},
        'Night': {'min_nurses': 1, 'min_senior': 1}
    },
    'Med-Surg': {
        'Morning': {'min_nurses': 3, 'min_senior': 2},
        'Evening': {'min_nurses': 2, 'min_senior': 1},
        'Night': {'min_nurses': 1, 'min_senior': 1}
    },
    'ICU': {
        'Morning': {'min_nurses': 2, 'min_senior': 1},
        'Evening': {'min_nurses': 1, 'min_senior': 1},
        'Night': {'min_nurses': 2, 'min_senior': 2}
    }
```

```
}

# Display requirements
print("\nSHIFT STAFFING REQUIREMENTS")
print("=" * 70)
for dept, shifts in shift_requirements.items():
    print(f"\n{dept} Department:")
    for shift, req in shifts.items():
        print(f"  {shift:8} - Min {req['min_nurses']} nurses (at least
{req['min_senior']} Senior/Expert)")

# Calculate total weekly demand
total_shifts_needed = 0
for dept in shift_requirements:
    for shift in shift_requirements[dept]:
        total_shifts_needed += shift_requirements[dept][shift]['min_nurses'] *
7  # 7 days

print("\n" + "=" * 70)
print(f"Total shift-slots needed per week: {total_shifts_needed}")
print(f"Available capacity (20 nurses × 6 shifts max): {20 * 6} shift-slots")
print(f"Capacity margin: {20 * 6 - total_shifts_needed} shift-slots")
```

```
SHIFT STAFFING REQUIREMENTS
======================================================================

ED Department:
  Morning  - Min 2 nurses (at least 1 Senior/Expert)
  Evening  - Min 2 nurses (at least 1 Senior/Expert)
  Night    - Min 1 nurses (at least 1 Senior/Expert)

Med-Surg Department:
  Morning  - Min 3 nurses (at least 2 Senior/Expert)
  Evening  - Min 2 nurses (at least 1 Senior/Expert)
  Night    - Min 1 nurses (at least 1 Senior/Expert)

ICU Department:
  Morning  - Min 2 nurses (at least 1 Senior/Expert)
  Evening  - Min 1 nurses (at least 1 Senior/Expert)
  Night    - Min 2 nurses (at least 2 Senior/Expert)

======================================================================
Total shift-slots needed per week: 112
Available capacity (20 nurses × 6 shifts max): 120 shift-slots
Capacity margin: 8 shift-slots
```

Week Structure

```
# Define the week structure
days_of_week = ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday',
'Saturday', 'Sunday']
shift_types = ['Morning', 'Evening', 'Night']

# Create a schedule template DataFrame
schedule_template = []
for day_num, day in enumerate(days_of_week):
    for shift in shift_types:
        is_weekend = day in ['Saturday', 'Sunday']
        schedule_template.append({
            'day': day,
            'day_num': day_num,
            'shift': shift,
            'is_weekend': is_weekend,
            'shift_code': f"{day[:3]}_{shift[:3]}"  # e.g., "Mon_Mor"
        })

schedule_template_df = pd.DataFrame(schedule_template)

print("\nWEEK SCHEDULE STRUCTURE (21 shifts total)")
print("=" * 60)
print(schedule_template_df.to_string(index=False))
```

```
WEEK SCHEDULE STRUCTURE (21 shifts total)
============================================================
       day  day_num   shift  is_weekend shift_code
    Monday        0 Morning       False    Mon_Mor
    Monday        0 Evening       False    Mon_Eve
    Monday        0   Night       False    Mon_Nig
   Tuesday        1 Morning       False    Tue_Mor
   Tuesday        1 Evening       False    Tue_Eve
   Tuesday        1   Night       False    Tue_Nig
 Wednesday        2 Morning       False    Wed_Mor
 Wednesday        2 Evening       False    Wed_Eve
 Wednesday        2   Night       False    Wed_Nig
  Thursday        3 Morning       False    Thu_Mor
  Thursday        3 Evening       False    Thu_Eve
  Thursday        3   Night       False    Thu_Nig
    Friday        4 Morning       False    Fri_Mor
    Friday        4 Evening       False    Fri_Eve
    Friday        4   Night       False    Fri_Nig
  Saturday        5 Morning        True    Sat_Mor
  Saturday        5 Evening        True    Sat_Eve
  Saturday        5   Night        True    Sat_Nig
```

```
   Sunday         6 Morning         True     Sun_Mor
   Sunday         6 Evening         True     Sun_Eve
   Sunday         6   Night         True     Sun_Nig
```

Minimal Helper Functions

We provide only basic data structures and a validation function. **You must implement the scheduling algorithm yourself.**

```
def validate_schedule(schedule, nurses_df, shift_requirements):
    """
    Validate a schedule against all constraints.

    Args:
        schedule: Dict or DataFrame representing nurse assignments
                  Format example: {('Monday', 'Morning', 'ED'): [nurse_id1,
nurse_id2, ...], ...}
        nurses_df: DataFrame with nurse information
        shift_requirements: Dict with staffing requirements

    Returns:
        dict with validation results and violations
    """
    violations = []
    warnings = []

    # This is a TEMPLATE - YOU need to implement the actual validation logic!
    # Check:
    # 1. All shifts have minimum staff
    # 2. Senior/Expert requirements met
    # 3. No nurse works more than 6 shifts per week
    # 4. No nurse works more than 5 consecutive days
    # 5. Minimum 11 hours rest between shifts
    # 6. Weekend distribution fairness

    print("Validation function is a template - you must implement full logic!")

    return {
        'is_valid': len(violations) == 0,
        'violations': violations,
        'warnings': warnings
    }

def calculate_schedule_cost(schedule, nurses_df):
    """
    Calculate total cost of a schedule.

    Args:
```

```
        schedule: Your schedule representation
        nurses_df: DataFrame with nurse information

    Returns:
        dict with cost breakdown
    """
    # Constants
    REGULAR_SHIFT_COST = 200
    OVERTIME_SHIFT_COST = 300  # 1.5x multiplier
    WEEKEND_PREMIUM = 50
    NIGHT_PREMIUM = 30
    UNDERSTAFFING_PENALTY = 500

    # YOU must implement this!
    # Calculate costs based on:
    # - Regular shifts vs overtime (>40 hours/week)
    # - Weekend premiums
    # - Night premiums
    # - Understaffing penalties

    print("Cost calculation function is a template - you must implement!")

    return {
        'total_cost': 0,
        'regular_cost': 0,
        'overtime_cost': 0,
        'premium_cost': 0,
        'penalty_cost': 0
    }

print("\nHelper function templates loaded")
print("YOU must implement the full logic!")
```

```
Helper function templates loaded
YOU must implement the full logic!
```

Your Task

You must develop a scheduling solution that creates a one-week schedule for all 20 nurses and provides:

1. Jupyter Notebook with Complete Solution

Your notebook should include:

- **Data exploration:** Insights about current staffing patterns and constraints
- **Algorithm implementation:** Your scheduling approach (construction + improvement)
- **Results:** Complete one-week schedule with all assignments

- **Fairness analysis:** Distribution of weekends, nights, workload across nurses
- **Cost analysis:** Total cost and breakdown
- **Robustness testing:** What-if analysis for 1-2 nurses calling in sick

> **Note**
> I won't judge the code quality in this notebook. It's just for me to review your final solution and identify any mistakes if your results seem unrealistic. The final project will be graded primarily based on your presentation and the results you have achieved.

2. Presentation

- **Problem understanding:** Dr. Torres's challenges in your own words
- **Your approach:**: Algorithm choice, how you handled constraints and fairness
- **Results:** Schedule visualization, cost savings, fairness metrics
- **Business impact:** Recommendations for NurseNext, implementation plan

> **Note**
> Please upload both, presentation and jupyter notebook (or `.py` files) for the final project.

3. Key Metrics to Report

- Total weekly cost
- Overtime cost reduction vs. simple greedy solution
- Fairness metrics:
  - Weekend distribution (mean, std, min, max)
  - Night shift distribution
  - Total shifts per nurse
- Constraint compliance (labor law violations: should ideally be ZERO!)
- Robustness: Impact of 1-2 sick nurses

Constraints and Requirements

> ! Important
> **Hard Constraints (Should Satisfy!)**
>
> 1. **Shift coverage:** All shifts meet minimum staffing requirements
> 2. **Skill requirements:** Each shift has required number of Senior/Expert nurses
> 3. **Maximum shifts:** No nurse works more than 6 shifts per week (48 hours)
> 4. **Maximum consecutive:** No nurse works more than 5 consecutive days
> 5. **Minimum rest:** At least 11 hours between shifts!
> 6. **Department qualification:** Nurses only work in departments they're qualified for
>   - **Specialists:** Can only work in their assigned department (check `departments_qualified`)
>   - **Flexible nurses:** Can work in any department (ED, Med-Surg, or ICU)
> 7. **Night shift eligibility:** Only nurses with `can_work_night=True` on night shifts
> 8. **Weekend monthly limit:** No nurse exceeds 2 weekend days per month
>   - Check `weekend_days_worked_this_month` field (shows days already worked in Weeks 1-2)
>   - If a nurse has already worked 2 weekend days, they **cannot** be assigned any weekend shifts this week
>
> **Soft Constraints (Optimize for Fairness)**
>
> 1. **Weekend fairness:** Equal distribution of weekend shifts
> 2. **Night shift fairness:** Equal distribution of night shifts among eligible nurses
> 3. **Workload balance:** Total shifts per nurse should be within ±1 shift of average
> 4. **Undesirable shift balance:** Fair distribution of "undesirable" shifts (nights, weekends)

Tips for Success

1. **Start with constraints:** Get a feasible schedule first, optimize fairness second
2. **Fairness is subjective:** Define clear metrics for "fair" distribution
3. **Visualize the schedule:** A Gantt chart or heat map makes patterns obvious
4. **Labor laws are non-negotiable:** Zero violations if possible. Check after every change!
5. **Think like a nurse:** What makes a schedule "good" from their perspective?
6. **Overtime is expensive but inevitable:** You have a structural staff shortage!

Common Pitfalls to Avoid

- **Ignoring labor laws:** Legal violations = instant failure in real world!
- **Unfair weekend distribution:** This is nurses' #1 complaint, track it carefully!
- **Not checking 11-hour rest:** Evening→Morning violates rest requirements
- **Forgetting skill requirements:** ICU night needs 2 Senior/Expert nurses!
- **Over-optimization:** A good, fair schedule beats a "perfect" but unfair one
- **No robustness testing:** Real hospitals have sick calls every week!

- **Poor visualization:** Make the schedule easy to read for nurses!

Data Access

All data is provided in this notebook:

- `nurses_df`: DataFrame with 20 nurses and their attributes
- `shift_requirements`: Dict with staffing requirements by department and shift
- `schedule_template_df`: Week structure (20 shifts)
- Helper templates: `validate_schedule()`, `calculate_schedule_cost()`

You must implement the full scheduling algorithm and constraint checking logic!

Deadline

- **Notebook submission & Presentation:** Lecture 12
- **Good luck, consultants!**

Bibliography